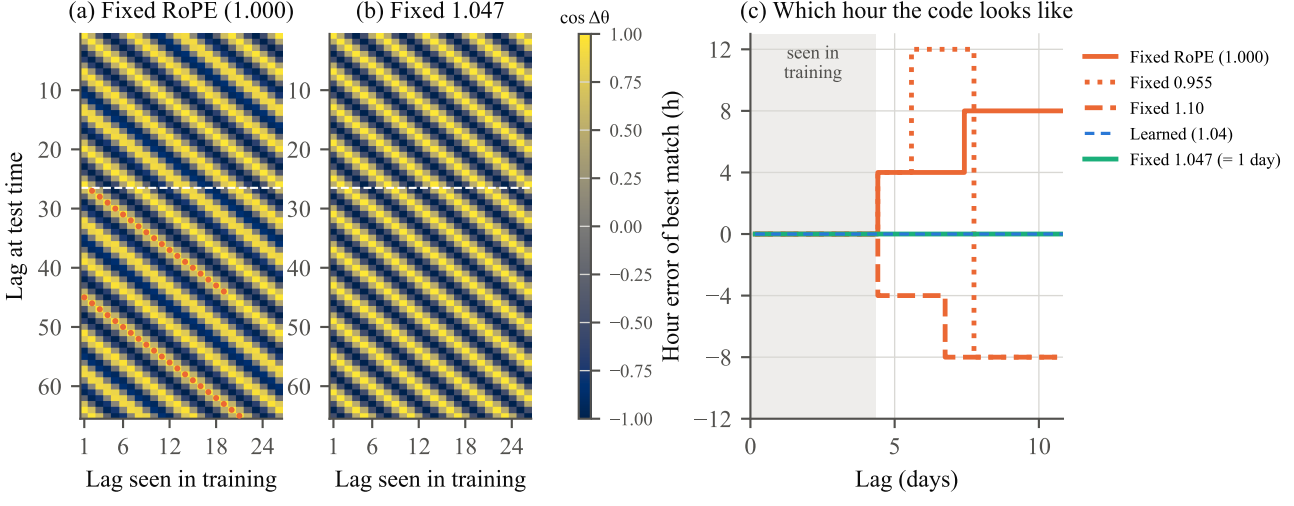
(a) Fixed RoPE (1.000)
(b) Fixed 1.047
cos Δθ
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Lag at test time
10
20
30
40
50
60
1
6
12
18
24
Lag seen in training
Lag seen in training
(c) Which hour the code looks like
seen in training
Hour error of best match (h)
12
8
4
0
−4
−8
−12
0
5
10
Lag (days)
Fixed RoPE (1.000)
Fixed 0.955
Fixed 1.10
Learned (1.04)
Fixed 1.047 (= 1 day)